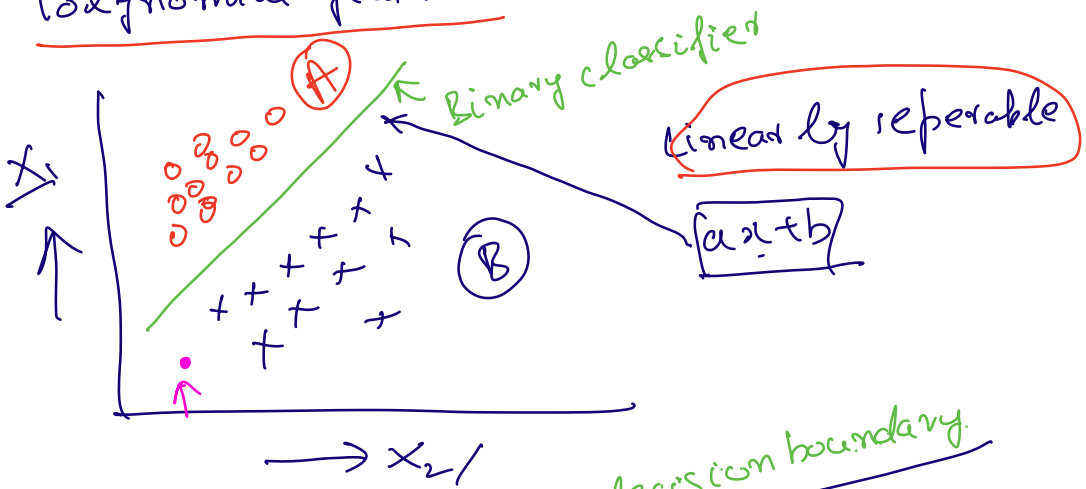
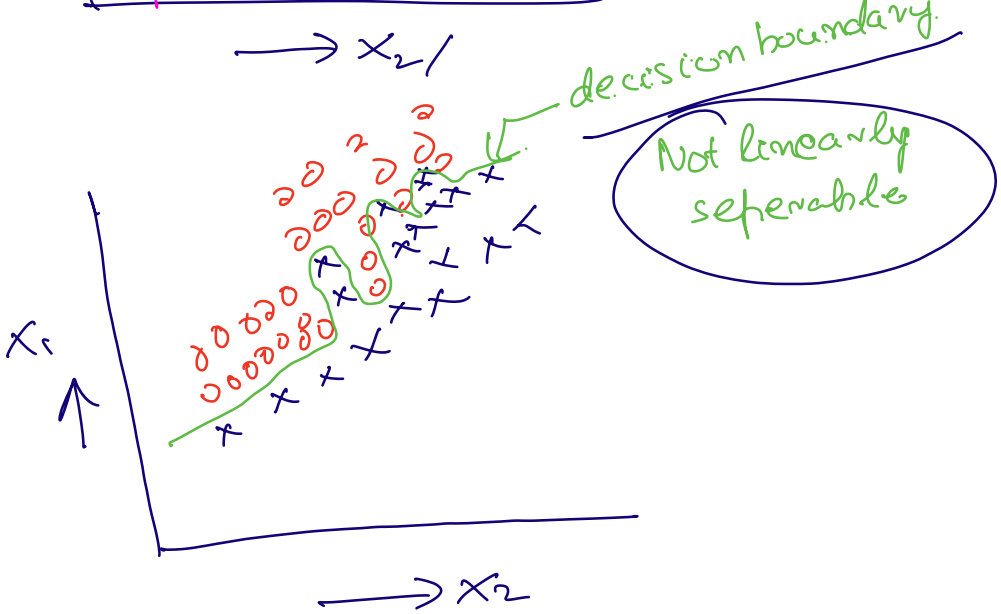
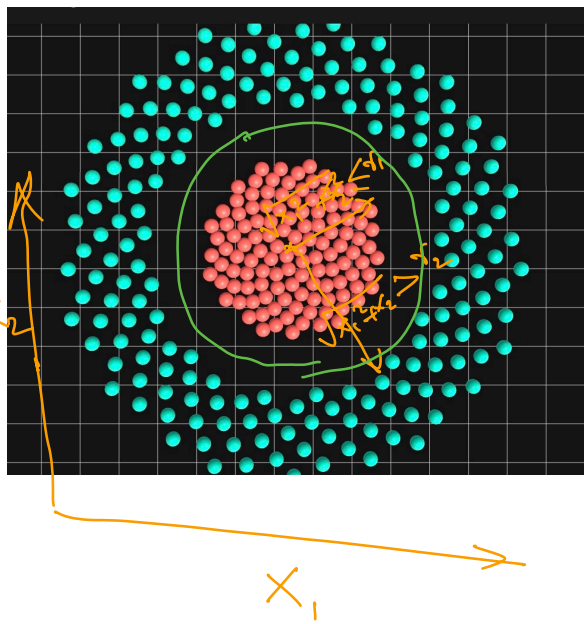
Support Vector Machine

1. Linear seperability ✓
2. Polynomial feature → ✓
3. Concept Geometric margin ✓
4. Linear Support Vector machine ✓
5. Kernel functions → Polynomial
   Gaussian — RBF

Polynomial features







$\phi(x_1, x_2) \rightarrow x_1, x_2, x_1^2 + x_2^2$

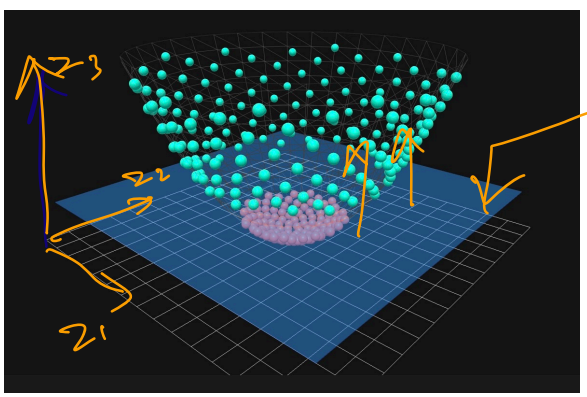
$t.(x_1^2 + x_2^2)$

$x_1 \quad x_2 \quad t$

$\uparrow \quad \uparrow \quad \uparrow$

z_1, z_2, z_3

$t \in [0, 1]$



plane

linearly seperable

optimal marginal classifier

$h_\theta(x) = g(\theta^T x)$ ← sampled set

h_θ ↑ o/p

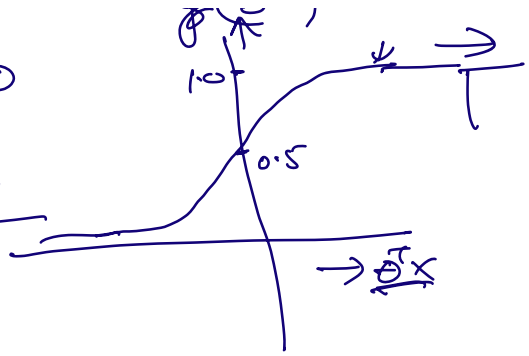
g ↑ sigmoid

θ → set of parameters

$e^{(\theta^T x)}$

If ($y^{(i)} = 1$) $\theta^T x >> 0$

If ($y^{(i)} = 0$) $\theta^T x << 0$



$y \in \{-1, +1\}$

$$g(z) = \begin{cases} 1, & \text{if } z \geq 0 \\ -1, & \text{if } z < 0 \end{cases}$$

$z = \theta^T x$

$$h_{w,b}(x) = g(w^T x + b)$$

bias

matrix $n \times m$

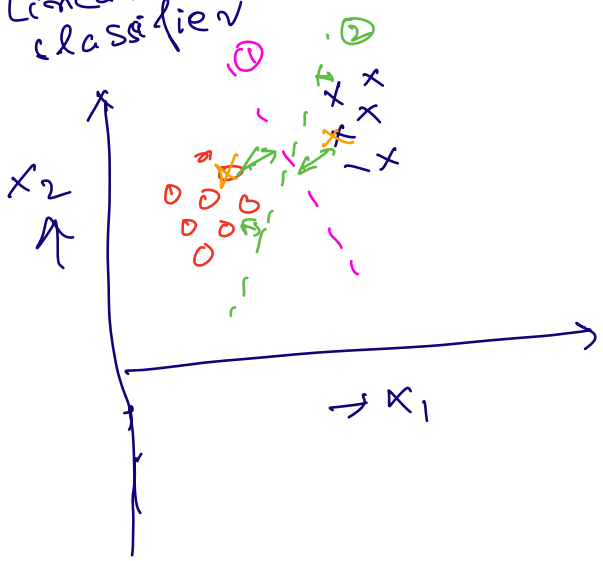
parameters

features examples

$$w^T x + b = \sum_{i=1} w_i x_i + b$$

$w_1 x_1^{(i)} + w_2 x_2^{(i)} + \cdots + w_n x_n^{(i)} + b$

Linear classifier



Functional margin

$$\hat{\gamma}^{(i)} = y^{(i)}(w^T x^{(i)} + b)$$

if $y^{(i)} = 1$, $w^T x + b \gg 0$

if $y^{(i)} = -1$, $w^T x + b \ll 0$

$$\hat{\gamma}_{min} = \min \hat{\gamma}^{(i)} \quad i = 1, 2, \ldots, m$$

$$\gamma^{(i)} = \frac{y^{(i)}(w^T x^{(i)} + b)}{\|w\| \leftarrow \sqrt{w_1^2 + w_2^2 + \ldots + w_n^2}}$$

$\gamma^{(i)}$ → geometric margin

$\|w\|$ → normalises

maximise $\left(\frac{y^{(i)}(w^T x^{(i)} + b)}{\|w\|} \right)_{min}$

$$\min_{w,b} \|w\|^2$$

$$s.t \quad y^{(i)}(w^T x^{(i)} + b) \geq 1$$

↑ convex optimisation

$$\frac{y^{(i)}(w^T x^{(i)} + b)}{\|w\|} \geq f_{min}$$

$$\|w\| = \frac{1}{f_{min}}$$

$$\max \frac{1}{\|w\|}$$

$$s.t \quad \frac{y^{(i)}(w^T x^{(i)} + b)}{\|w\|} \geq f$$

$$s.t \quad y^{(i)}(w^T x^{(i)} + b) \geq 1$$

$$\min_{w,b} \frac{1}{2}\|w\|^2$$

$$s.t \quad y^{(i)}(w^T x^{(i)} + b) \geq 1$$

$$i = 1, 2, \cdots, m$$

objective

Representer Theorem

$$w = \sum_{i=1}^{m} \alpha_i x^{(i)}$$

$$\rightarrow w = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_m x_m$$

$$\|w\|^2 = w^T w$$

$$\min \frac{1}{2} w^T w = \frac{1}{2} \left(\sum_{i=1}^{m} \alpha_i y^{(i)} x^{(i)}\right)^T \left(\sum_{j=1}^{m} \alpha_j y^{(j)} x^{(j)}\right)$$

$$s.t \quad y^{(i)} \sum_{k=1}^{m} \alpha_k \cdot y^{k} x^{(k)} + b \geq 1$$

$$\max \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i} \sum_{j} y^{(i)} y^{(j)} \alpha_i \alpha_j \underbrace{x^{(i)T} x^{(j)}}_{\text{inner product}}$$

$$s.t \quad \alpha_i \geq 0$$

$$\& \sum_{k=1}^{m} y^{(k)} x^{(k)} = 0$$

$$\max \sum_{i} \alpha_i - \frac{1}{2} \sum_{i} \sum_{j} y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle$$

$$s.t \quad \alpha_i \geq 0$$

$$\& \sum_{k=1}^{m} y^{(k)} x^{(k)} = 0$$

$X \rightarrow \phi(X) \rightarrow$ higher dimensional feature

$X_1, X_2 \rightarrow \phi(X) \rightarrow$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_1 x_2 \\ x_1^2 \\ x_2^2 \\ x_1^2 x_2 \\ x_2^2 x_1 \\ x_1^3 \\ x_2^3 \\ \vdots \end{bmatrix}$$

2D $\rightarrow$ 1M

$K(x^{(i)}, x^{(j)}) = \phi(x^{(i)})^T \phi(x^{(j)})$

↑

w/o doing the transformation we are able compute the inner product of the higher dimension feature space

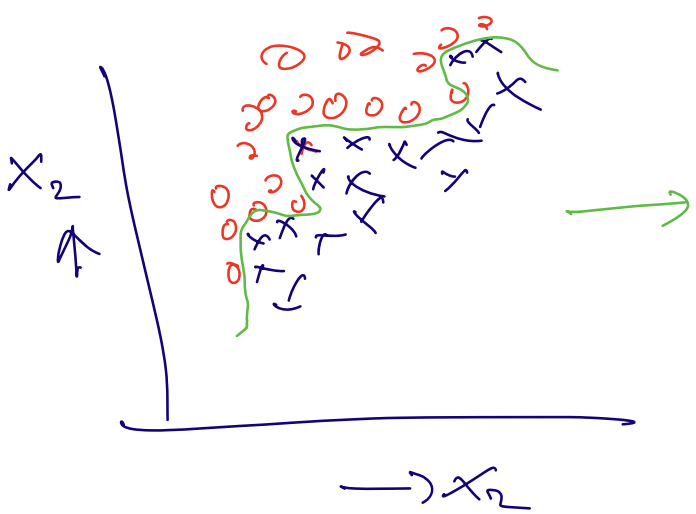
Polynomial kernel function

1 x_1, x_2, x_3

d=5 $x_1^2, x_2^2 \ldots\ldots x_1^5, x_2^5, x_1 x_2^2 x_3^3$

$$K(x, z) = (x^T z + c)^d \quad c \in \mathbb{R}$$

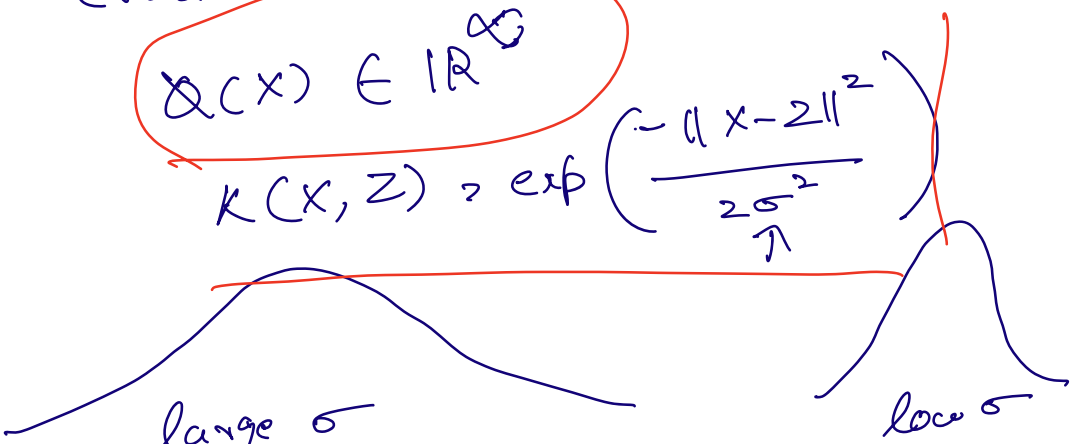
$x^{(i)}$ $x^{(j)}$



Gaussian Kernel Function
(Radial Basis Function)

$\phi(x) \in \mathbb{R}^{\infty}$

$K(x, z) = \exp\left(\frac{-\|x-z\|^2}{2\sigma^2}\right)$



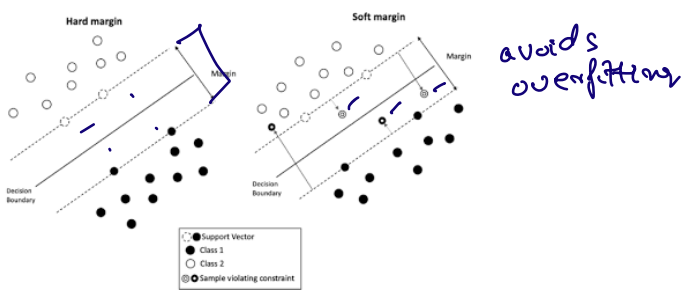
large σ
high bias
low variance

low σ
low bias
high variance
overfitting

Hard margin & soft margin

$$\min \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{m} \xi_{(i)} \quad \xi \leftarrow \text{error}$$

$$\text{s.t } y^{(i)}(w^T x^{(i)} + b) \geq 1 - \xi_i \quad i = 1, 2, \ldots, m$$



avoids overfitting